

Velocity Corporate Training Center, Pune

Deep Learning Notes

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1. Introduction to Deep Learning

Deep Learning is a subset of **Machine Learning (ML)** that focuses on algorithms modeled after the human brain's structure, known as **Artificial Neural Networks (ANNs)**. These networks consist of many layers of neurons, which is why deep learning models are also called **deep neural networks**. The depth (i.e., number of layers) allows them to learn hierarchical patterns, making them especially powerful for high-dimensional and unstructured data like images, audio, and natural language.

Why Deep Learning?

While traditional ML models (like decision trees or logistic regression) require handcrafted features, deep learning models learn these features automatically, layer by layer. This automatic feature learning makes deep learning superior in domains like:

- **Image and Video Processing:** Automatically identifying objects or faces
- **Natural Language Processing (NLP):** Understanding human language
- **Time Series Forecasting:** Predicting stock prices or weather patterns

Real-World Applications

- **Computer Vision:** Object detection in autonomous vehicles, facial recognition
- **NLP:** Machine translation, voice assistants, chatbots
- **Healthcare:** Diagnosing diseases from medical images
- **Finance:** Credit scoring, fraud detection
- **Agriculture:** Detecting crop diseases using drone images

Classification vs. Regression

- **Classification:** Predicts discrete labels or categories (e.g., spam or not spam)
- **Regression:** Predicts continuous values (e.g., house price)

Deep learning can solve both classification and regression problems using the same underlying architecture but with different output layers and loss functions.

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2. Model Lifecycle in Deep Learning Projects

1. **Problem Definition:** Define the task — classification, regression, etc.
2. **Data Collection:** Gather raw data from sensors, web, APIs, etc.
3. **Data Preprocessing:** Cleaning, missing value handling, normalization, encoding.
4. **Train-Validation-Test Split:** Typically 70-15-15 or 80-10-10.
5. **Model Design:** Choose network architecture (e.g., CNN, LSTM, Transformer).
6. **Training:** Fit the model to training data using appropriate loss and optimizer.
7. **Evaluation:** Assess using metrics like accuracy or RMSE.
8. **Hyperparameter Tuning:** Optimize architecture and training parameters.

9. **Deployment:** Convert the model into a service (e.g., Flask, FastAPI, TensorFlow Serving).
10. **Monitoring & Retraining:** Track real-world performance and update when needed.

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3. Neural Network Fundamentals

A **neural network** is a computational system composed of interconnected processing units called **neurons** or **nodes**. These neurons are arranged in layers and connected by weighted links.

Key Concepts and Terminologies

Term	Description	Example
Input Features (x1, x2, ..., xn)	Independent variables or data points fed into the network	Age, salary, temperature
Neuron (Node)	Fundamental unit that performs computation	Mimics a biological neuron
Weights (w1, w2, ..., wn)	Coefficients applied to input features to control their influence on the output	High weight = high influence
Bias (b)	Constant term added to the weighted sum to shift the activation threshold	Allows model to shift decision boundary
Activation Function	Introduces non-linearity to help learn complex patterns	ReLU, Sigmoid, Tanh, etc.
Output	Final prediction of the model, varies based on task type (classification or regression)	0 or 1 (classification), numerical value (regression)

Network Architecture

- Input Layer:** Takes raw features as input.
- Hidden Layers:** Intermediate layers that apply transformations to extract patterns.
- Output Layer:** Produces prediction; for classification, uses activation like Sigmoid or Softmax, and for regression, uses Linear activation.

Example (Classification):

Task: Email spam detection

- Input features: Frequency of words like “win”, “free”
- Hidden layers: Combine features to detect spam signals
- Output: 0 (not spam) or 1 (spam) via **Sigmoid activation**

Example (Regression):

Task: Predict house price

- Input features: Size, location, number of rooms
- Hidden layers: Learn interactions like “location × size”
- Output: Continuous value like ₹50,00,000 using **Linear activation**

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4. How Neural Networks Learn

Training a neural network means adjusting its weights and biases such that its predictions are close to the true outputs. This process is iterative and is typically done using **gradient descent** based optimization.

Detailed Training Process

1. **Weight Initialization:**

- Initialize all weights and biases with small random numbers.

2. **Forward Propagation:**

- Data flows from input layer to output layer.
- At each neuron:
 - $z = w1 \times x1 + w2 \times x2 + \dots + wn \times xn + b$
 - Apply activation: $a = \text{activation}(z)$

3. **Loss Calculation:**

- Measure the prediction error.

- Common loss functions:
 - Mean Squared Error (MSE):** $1/N * \sum (y_{true} - y_{pred})^2$ (used in regression)
 - Binary Cross-Entropy:** $-y*\log(p) - (1-y)*\log(1-p)$ (binary classification)
 - Categorical Cross-Entropy:** $-\sum(y_i*\log(p_i))$ (multi-class classification)

4. **Backpropagation:**

- Use **Chain Rule** to compute the gradient of the loss with respect to each weight.
- These gradients tell us how to adjust the weights to minimize loss.

5. **Optimizer (Weight Update):**

- Update weights using algorithms like:
 - Stochastic Gradient Descent (SGD)**
 - Adam (Adaptive Moment Estimation)**
- Update rule: $w_{new} = w_{old} - \eta * \partial Loss / \partial w$
 - η (eta) is the **learning rate**

6. **Epoch and Iterations:**

- Epoch:** One full pass through the entire training data
- Batch:** Subset of data used in each update step
- Training continues for many epochs until loss converges

 Goal:

To minimize the loss function and improve model generalization on unseen data.

 Example (Binary Classification):

Problem: Predict whether a customer will default on a loan

- Target: 0 or 1
- Loss Function: Binary Cross-Entropy
- Final layer: Sigmoid activation

 Example (Regression):

Problem: Predict temperature based on humidity and wind speed

- Target: Continuous temperature value
- Loss Function: MSE
- Final layer: Linear activation

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5. Weight Initialization in Deep Learning

Weight initialization sets the starting values of neural network weights before training. Proper initialization is critical for:

- Breaking symmetry so neurons learn different features.
- Preventing vanishing or exploding gradients.
- Accelerating convergence and improving model performance.

Common Problems Without Proper Initialization

- Vanishing Gradients:** Gradients become too small, stopping learning in early layers.
- Exploding Gradients:** Gradients become too large, causing unstable training.
- Slow or No Convergence:** Training stalls or becomes very slow.

Important Weight Initialization Techniques

1. **Xavier Initialization (Glorot Initialization)**

- Balances variance of activations and gradients by considering both input and output layer sizes.
- Formula (uniform):

$$W \sim U \left(-\sqrt{\frac{6}{n_{in} + n_{out}}}, \sqrt{\frac{6}{n_{in} + n_{out}}} \right)$$

- Best for sigmoid and tanh activations.
- Example: For 20 inputs and 10 outputs, weights initialized from $[-0.447, 0.447]$.
- **Advantages:** Stabilizes forward and backward signals; reduces vanishing/exploding gradients.
- **Disadvantages:** Less effective with ReLU activations.

2. He Initialization (Kaiming Initialization)

- Designed for ReLU and its variants; uses only the number of input neurons.
- Formula (uniform):

$$W \sim U\left(-\sqrt{\frac{6}{n_{in}}}, \sqrt{\frac{6}{n_{in}}}\right)$$

- Example: For 20 inputs, weights initialized from $[-0.548, 0.548]$.
- **Advantages:** Maintains variance after ReLU activations; speeds up convergence.
- **Disadvantages:** Not ideal for sigmoid/tanh activations.

3. Orthogonal Initialization

- Initializes weights as orthogonal matrices, preserving input norm.
- Useful for very deep networks and RNNs.
- **Advantages:** Prevents gradient issues; stabilizes training.
- **Disadvantages:** Computationally expensive; less common for simple feedforward nets.

Other Methods (Briefly)

- **Random Initialization:** Weights chosen randomly without scaling; can cause unstable training.
- **Zero Initialization:** Causes symmetry and no learning; not recommended.
- **Lecun Initialization:** For SELU activations.

Key Takeaways

- Initialization ranges depend on layer sizes and activation functions.
- Weights are usually initialized **symmetrically around zero**, not just between 0 and 1.
- Proper initialization ensures stable signal flow, effective learning, and faster convergence.
- Modern networks typically use **He initialization** for ReLU and **Xavier initialization** for sigmoid/tanh.

Tuning of Weight Initialization:

```
initializer_1 = hp.Choice('kernel_initializer_1', ['he_normal', 'glorot_uniform', 'random_normal'])
```

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6. Activation Functions

Activation functions are crucial for introducing **non-linearities** into the network. Without them, even a multi-layer neural network would behave like a simple linear model.

Common Activation Functions

Function	Formula	Range	Characteristics	Common Use Cases
Sigmoid (Logistic)	$1 / (1 + e^{-x})$	(0, 1)	Output interpretable as probability; used for binary classification	Output layer in binary classification models
Tanh (Hyperbolic Tangent)	$(e^x - e^{-x}) / (e^x + e^{-x})$	(-1, 1)	Zero-centered; better gradient flow than sigmoid	Hidden layers in RNNs, LSTMs
ReLU (Rectified Linear Unit)	$\max(0, x)$	$[0, \infty)$	Fast and sparse; can lead to dead neurons	Default for most hidden layers
Leaky ReLU	x if $x > 0$ else αx	$(-\infty, \infty)$	Avoids dead neurons by allowing small negative slope	Deep architectures, GANs
PReLU (Parametric ReLU)	x if $x > 0$ else ax	$(-\infty, \infty)$	Trainable slope on negative side	Advanced models requiring flexibility

Function	Formula	Range	Characteristics	Common Use Cases
ELU (Exponential Linear Unit)	x if $x > 0$ else $\alpha(e^x - 1)$	$(-\alpha, \infty)$	Smooth gradients and faster learning	Very deep networks
SELU (Scaled ELU)	$\lambda * \text{ELU}(x)$	$(-\infty, \infty)$	Self-normalizing; stable deep networks	Deep nets with many layers
Softmax	$e^{(x_i)} / \sum e^{(x_j)}$	$(0, 1)$, sum=1	Converts vector to probability distribution	Output layer in multi-class classification

Activation for Output Layer by Task

Problem Type	Output Activation Function
Binary Classification	Sigmoid
Multi-class Classification	Softmax
Regression	Linear (Identity)

Example:

- **Classification:** MNIST digit classification
 - Input: 28×28 pixel image
 - Output: Softmax layer with 10 neurons (0–9 digits)
- **Regression:** Predicting fuel efficiency
 - Input: Engine size, weight
 - Output: Linear activation (real number prediction)

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7. Optimizers in Deep Learning

An **optimizer** is an algorithm or mathematical method used in training neural networks, responsible for updating the weights and biases of the network to minimize the loss function and enhance model performance.

Role of Optimizers:

- Adjust model parameters efficiently.
- Ensure convergence to optimal solutions.
- Improve accuracy and stability of neural networks.

Importance of Optimizers

- Control the learning speed and direction of weight updates.
- Reduce oscillations and stabilize learning.
- Avoid getting trapped in suboptimal local minima.
- Enhance generalization performance.

Types of Optimizers

Below are the most important and widely used optimizers in Deep Learning:

1. Stochastic Gradient Descent (SGD)

Description: SGD updates weights incrementally using the gradient calculated from individual training examples or small batches of data. It frequently updates parameters, leading to faster initial convergence.

Formula:

$$W = W - \eta \frac{\partial L}{\partial W}$$

- W : Weights
- η : Learning rate
- $\frac{\partial L}{\partial W}$: Gradient of loss L w.r.t. weights W

Advantages:

- Simple and computationally efficient.
- Fast initial training speed.

- Memory efficient; suitable for large datasets.

Disadvantages:

- Can cause noisy and unstable updates.
 - High variance in weight updates.
-

2. SGD with Momentum

Description: SGD with Momentum uses a momentum term (velocity) to accelerate updates in relevant directions, reducing oscillations and smoothing the path to convergence.

Formula:

$$v_t = \gamma v_{t-1} + \eta \frac{\partial L}{\partial W}, \quad W = W - v_t$$

- v_t : velocity (momentum term)
- γ : momentum factor (usually 0.9)

Advantages:

- Reduces oscillations.
- Faster and more stable convergence.

Disadvantages:

- Additional hyperparameter tuning required (γ).
-

3. RMSProp (Root Mean Square Propagation)

Description: RMSProp adapts the learning rate for each parameter by maintaining an exponentially decaying average of squared gradients, helping to overcome vanishing or exploding gradients.

Formula:

$$E[g^2]_t = \gamma E[g^2]_{t-1} + (1 - \gamma) \left(\frac{\partial L}{\partial W} \right)^2$$
$$W = W - \frac{\eta}{\sqrt{E[g^2]_t + \epsilon}} \frac{\partial L}{\partial W}$$

- γ : decay rate (~ 0.9)
- ϵ : small value ($\sim 10^{-8}$) to avoid division by zero

Advantages:

- Adaptive learning rates help handle gradient instability.
- Effective for recurrent neural networks (RNNs/LSTMs).

Disadvantages:

- Sensitive to learning rate and decay rate settings.
-

4. Adam (Adaptive Moment Estimation)

Description: Adam combines the benefits of RMSProp (adaptive learning rate) and Momentum (directional acceleration). It computes individual adaptive learning rates for different parameters.

Formula:

- **First moment (mean gradients):**

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \frac{\partial L}{\partial W}$$

- **Second moment (mean squared gradients):**

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) \left(\frac{\partial L}{\partial W} \right)^2$$

- **Bias-corrected moments:**

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

- **Weight update:**

$$W = W - \frac{\eta}{\sqrt{\hat{v}_t + \epsilon}} \hat{m}_t$$

- Default values: $\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$

Advantages:

- Quick and efficient convergence.
- Automatically adaptive learning rates.
- Low sensitivity to hyperparameters.

Disadvantages:

- Sometimes may converge prematurely or overshoot optimal minima.

5. AdamW (Adam with Weight Decay)

Description: AdamW modifies Adam by explicitly incorporating weight decay (L2 regularization). This approach improves regularization and generalization, particularly in large neural networks.

Formula:

- Standard Adam update with additional weight decay term:

$$W = W - \eta \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}} + \lambda W \right)$$

- λ : weight decay parameter (regularization strength)

Advantages:

- Enhances generalization and reduces overfitting.
- Widely used for training Transformer models (like BERT and GPT).

Disadvantages:

- Introduces an additional hyperparameter (weight decay strength).

6. Nadam (Nesterov Accelerated Adam)

Description: Nadam combines Adam’s adaptive moment estimation with Nesterov accelerated momentum. It adjusts parameters considering the gradient at the anticipated future position, ensuring more precise updates.

Formula:

- Uses Nesterov’s look-ahead gradient:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \frac{\partial L}{\partial W}, \quad \hat{m}_t = \frac{m_t}{1 - \beta_1^t}$$

- Weight updates incorporating Nesterov momentum:

$$W = W - \frac{\eta}{\sqrt{\hat{v}_t + \epsilon}} \left(\beta_1 \hat{m}_t + (1 - \beta_1) \frac{\partial L}{\partial W} \right)$$

Advantages:

- Stable and accurate convergence.
- Often outperforms Adam on challenging problems.

Disadvantages:

- Slightly more computationally intensive.

📌 Quick Reference for Choosing Optimizers

Optimizer	Stability	Convergence Speed	Applications
SGD	Moderate	Moderate	Basic models, linear/logistic regression
SGD + Momentum	High	Moderate-Fast	CNNs, stable and faster convergence
RMSProp	High	Moderate-Fast	RNN/LSTM, time-series data

Optimizer	Stability	Convergence Speed	Applications
Adam (default)	High	Fast	General-purpose, CNN, NLP, most scenarios
AdamW	High	Fast	Large-scale models, NLP transformers (BERT/GPT)
Nadam	High	Fastest	Precision-critical scenarios

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8. Overfitting in Deep Learning

Overfitting occurs when a neural network learns not only the underlying patterns but also the noise in the training data. As a result, the model performs exceptionally well on training data but poorly on unseen (test) data, causing poor generalization.

In simpler terms:

- **Model memorizes** rather than learns patterns.
- Great accuracy on training data but low accuracy on test data.

Symptoms of Overfitting:

- High training accuracy with low validation/test accuracy.
- A large gap between training and validation loss.
- Poor model performance when deployed on new data.

Techniques to Avoid Overfitting:

1. Regularization (L1 & L2)
2. Early Stopping
3. Dropout
4. Batch Normalization
5. Reducing Model Complexity
6. Data Augmentation

Best Practices:

- Always **split data** into training, validation, and test sets.
- Employ **early stopping** to halt unnecessary training.
- Use **regularization techniques (L1/L2, Dropout)**.
- Consider **data augmentation** (especially for image/speech tasks).
- Apply **batch normalization** to stabilize training.
- Adjust **model complexity** based on the size and complexity of the dataset.

Implementing a combination of these strategies often yields the best results in preventing overfitting and ensuring robust, generalizable deep learning models.

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8.1 Regularization (L1 & L2)

- **Definition:** Adds a penalty to the loss function based on the magnitude of weights.
- **Types:**
 - **L1 Regularization (Lasso):** Penalizes absolute value of weights, leading to sparse weight matrices.
 - **L2 Regularization (Ridge):** Penalizes square of weights, forcing weights toward smaller values without making them zero.
- **Advantages:**
 - Helps generalize better.
 - Reduces model complexity.
- **Disadvantages:**
 - Excessive regularization can lead to underfitting.

Example (using Keras):

```
from tensorflow.keras.layers import Dense
from tensorflow.keras.regularizers import l2

model.add(Dense(128, activation='relu', kernel_regularizer=l2(0.01)))
```


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8.2 Early Stopping

- Early Stopping is a regularization technique used during training of neural networks to prevent overfitting by stopping training before the model starts to memorize the training data
- **How it works:**
 - During training, the model's performance is evaluated on a **validation dataset** after every epoch.
 - If the validation loss **does not improve** for a specified number of consecutive epochs (called **patience**), training is stopped.
 - The model weights from the epoch with the **best validation performance** are restored.
- **Advantages:**
 - Prevents wasting time training beyond the point of generalization.
 - Stops overfitting by not allowing the model to learn noise or irrelevant details.
 - Helps select the model that performs best on unseen data..
- **Disadvantages:**
 - Requires careful tuning of the patience parameter.

```
from tensorflow.keras.callbacks import EarlyStopping
early_stop = EarlyStopping(monitor='val_loss', patience=3, restore_best_weights=True)
model.fit(x_train, y_train, validation_data=(x_val, y_val), epochs=100, callbacks=[early_stop])
```

Example Use in Keras

```
from tensorflow.keras.callbacks import EarlyStopping

early_stop = EarlyStopping(monitor='val_loss', patience=3, restore_best_weights=True)

model.fit(x_train, y_train, validation_data=(x_val, y_val), epochs=100, callbacks=[early_stop])
```

Summary Table

Aspect	Description
Purpose	Prevent overfitting by stopping training early
Trigger	No improvement in validation metric for set patience epochs
Benefit	Saves time, improves generalization
Commonly Used With	Validation loss or accuracy metrics

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8.3 Dropout

- Dropout is a regularization technique to **prevent overfitting** in neural networks.
- During **training**, it randomly sets a fraction of neuron outputs to zero.
- This forces the network to learn **robust, distributed features**.
- During **inference**, dropout is turned off; all neurons are used.
- **Advantages:**
 - Reduces overfitting by preventing co-adaptation of neurons.
 - Encourages redundant, robust feature learning.
 - Easy to implement and widely effective.
- **Disadvantages:**
 - Can increase training time slightly.
 - Might require careful tuning.

Example:

```
model.add(Dropout(dropout_1))
```

Parameter	Value
Number of neurons	5
Outputs before dropout	[0.8, 0.1, 0.7, 0.3, 0.5]
Dropout rate	0.4 (40%)

Scale Factor Calculation

- Dropout rate $p = 0.4$
- Scale factor = $\frac{1}{1-p} = \frac{1}{0.6} \approx 1.67$ This scale factor compensates for the dropped neurons so that the expected sum of activations remains consistent.

Training Phase

- Randomly drop 40% of neurons; suppose neurons 2 and 5 are dropped.

Neuron	Output Before Dropout	Output After Dropout	Scaled Output (Output ÷ 0.6)
1	0.8	0.8	1.336
2	0.1	0	0
3	0.7	0.7	1.167
4	0.3	0.3	0.5
5	0.5	0	0

Inference Phase

- No neurons are dropped; all outputs are used as is:

Neuron	Output
1	0.8
2	0.1
3	0.7
4	0.3
5	0.5

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4. Batch Normalization

- Normalizes inputs of each layer **per mini-batch** to zero mean and unit variance.
- Applies **learnable scale (γ) and shift (β)** after normalization.
- Reduces **internal covariate shift**, stabilizes and speeds up training.

Why Use It?

- Enables higher learning rates.
- Improves training stability and gradient flow.
- Acts as a mild regularizer.

How It Works (Simplified)

$$\hat{x}_i = \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}, \quad y_i = \gamma \hat{x}_i + \beta$$

- μ_B, σ_B^2 : mean and variance of the mini-batch.
- ϵ : small constant for numerical stability.

Example

Sample	Input		Normalized	Scaled & Shifted $y_i = \gamma \hat{x}_i + \beta$
	x_i	$\hat{x}_i = \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}$		
1	4.0	$\frac{4.0 - 7.0}{\sqrt{5 + \epsilon}} = -1.342$		$2 \times (-1.342) + 1 = -1.684$
2	6.0	$\frac{6.0 - 7.0}{\sqrt{5 + \epsilon}} = -0.447$		$2 \times (-0.447) + 1 = 0.106$

Sample	Input x_i	Normalized		Scaled & Shifted $y_i = \gamma \hat{x}_i + \beta$
		$\hat{x}_i$	$= \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}$	
3	8.0	$\frac{8.0 - 7.0}{\sqrt{5 + \epsilon}}$	$= 0.447$	$2 \times 0.447 + 1$ $= 1.894$
4	10.0	$\frac{10.0 - 7.0}{\sqrt{5 + \epsilon}}$	$= 1.342$	$2 \times 1.342 + 1$ $= 3.684$

Example (using Keras):**

```
from tensorflow.keras.layers import BatchNormalization, Dense
```

```
model.add(Dense(128, activation='relu'))
model.add(BatchNormalization())
```

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5. Reducing Model Complexity

- **Definition:** Reducing the number of layers or neurons.
- **Advantages:**
 - Simpler models generalize better.
 - Reduced training and inference time.
- **Disadvantages:**
 - Risk of underfitting if the model becomes too simple.

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6. Data Augmentation

- **Definition:** Artificially expanding the dataset by applying transformations (e.g., rotation, scaling, cropping).
- **Common in:** Image data, speech recognition.
- **Advantages:**
 - Increases data diversity.
 - Improves generalization capability.
- **Disadvantages:**
 - Might not be applicable or intuitive for some datasets (e.g., tabular data).

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9. Gradient Problem in NN

1. Vanishing Gradient Problem

Definition:

In deep neural networks, **Vanishing Gradient** refers to a condition where the gradients (derivatives of the loss function with respect to the weights) become extremely small as they propagate backwards through the network layers. When gradients approach zero, updates to weights in earlier layers become negligible, causing the model to stop effectively learning new features.

Why it Happens:

- Repeated multiplications of small values in deep networks.
- Usage of activation functions with small gradients, like sigmoid or tanh, especially in deeper networks.

Example:

Consider a neural network with multiple hidden layers, each using a sigmoid activation function:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

The derivative of the sigmoid activation function is:

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$

Because $\sigma(x)$ outputs values between 0 and 1, the derivative $\sigma'(x)$ will be between 0 and 0.25. Multiplying many such small values when propagating gradients backward through multiple layers drastically reduces the gradient magnitude.

For instance, consider:

- Initial gradient at output layer: 0.1
- Passing through 10 layers with sigmoid activation (each gradient multiplied by approx. 0.1):

$$\text{Gradient at layer 1} \approx 0.1 \times 0.1 \times 0.1 \times \dots (10 \text{ times}) = 0.1^{10} = 10^{-10}$$

- Such tiny gradients mean no significant weight updates, essentially halting learning.

Consequences:

- Early layers learn slowly or not at all.
- Poor convergence or suboptimal performance.
- Network fails to capture complex features in the data.

Solutions:

- Use activation functions with less vanishing gradient issues (e.g., **ReLU**, **Leaky ReLU**).
- Techniques like **Batch Normalization** and **Skip Connections (ResNet)** help maintain stable gradients.
- Appropriate **weight initialization** methods (**Xavier/Glorot initialization**, **He initialization**) help mitigate this problem.

2. Exploding Gradient Problem

Definition:

The **Exploding Gradient Problem** occurs when gradients become excessively large during backpropagation, leading to large updates in weights. This typically causes numerical instability and divergence during training.

Why it Happens:

- Multiplying several large gradients together, especially common in Recurrent Neural Networks (RNNs).
- Poor choice of learning rate or unstable network structure.

Example:

Consider an example where gradients are consistently greater than 1 during backpropagation:

- Suppose each gradient at each layer is approximately 3. If we have 10 layers:

$$\text{Gradient at layer 1} \approx 3 \times 3 \times 3 \dots (10 \text{ times}) = 3^{10} = 59,049$$

- This excessively large gradient (59,049) will cause massive updates to the weights, making the training unstable.

Consequences:

- Weight values become excessively large or small, causing instability.
- Divergence of loss function during training.
- Training becomes highly unstable, leading to NaN values in loss and model failure.

Solutions:

- **Gradient Clipping**: Limit gradient values within a threshold. E.g., TensorFlow implementation:

```
optimizer = tf.keras.optimizers.Adam(clipnorm=1.0)
```

- Lower learning rate or adaptive optimizers like Adam, RMSProp.
- Initialization strategies to prevent large initial weights (Xavier/Glorot initialization).
- Batch Normalization to maintain gradient stability.

Summary Comparison

Aspect	Vanishing Gradient	Exploding Gradient
Gradient Size	Very small (approaches zero)	Extremely large (huge values)
Typical Cause	Deep nets, sigmoid/tanh	Deep nets, unstable training
Effect	Slow or halted training	Training instability/diverge
Common Solution	ReLU, Batch Norm, Residual Connections	Gradient Clipping, Batch Norm

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10. Model Evaluation Metrics

Once a deep learning model is trained, it’s essential to evaluate how well it performs — not just on training data, but more importantly on unseen validation and test data. Evaluation metrics provide quantitative measures to assess a model’s accuracy, precision, recall, or error, depending on the task type.

Evaluation for Classification Tasks

Metric	Description	Use Case
Accuracy	The proportion of correct predictions out of all predictions: $\frac{TP + TN}{TP + TN + FP + FN}$	When classes are balanced
Precision	The proportion of true positive predictions out of all predicted positives: $\frac{TP}{TP + FP}$	When False Positives are costly (e.g., spam detection)
Recall	The proportion of actual positives correctly predicted: $\frac{TP}{TP + FN}$	When False Negatives are critical (e.g., cancer diagnosis)
F1 Score	Harmonic mean of precision and recall: $2 * \frac{Precision * Recall}{Precision + Recall}$	When you want balance between precision and recall
ROC-AUC	Area under the ROC Curve, shows ability to distinguish classes	Binary classification (e.g., fraud detection)
Confusion Matrix	Matrix summarizing TP, TN, FP, FN	Analyzing classification errors

Evaluation for Regression Tasks

Metric	Description	Use Case
Mean Absolute Error (MAE)	Average absolute difference between predicted and actual values	When all errors are equally weighted
Mean Squared Error (MSE)	Average of squared differences; penalizes larger errors more	Regression when large errors must be penalized
Root Mean Squared Error (RMSE)	Square root of MSE	Similar to MSE but in original units
R² Score	Proportion of variance explained by the model ($1 - \frac{SS_{res}}{SS_{tot}}$)	Indicates how well the model explains the data

Example (Classification):

Predicting if an email is spam or not:

- Precision: What fraction of predicted spam emails are actually spam?
- Recall: What fraction of all spam emails did the model detect?

Example (Regression):

Predicting house prices:

- MSE penalizes large errors more than MAE.
- R² Score tells you how much variance in price is explained by the model.

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11. Hyperparameter Tuning in Deep Learning

Hyperparameters are settings defined before training a model that guide the learning process. Unlike model parameters (weights and biases), hyperparameters are not learned — they are manually or automatically selected.

Common Hyperparameters

- **Learning Rate (η):** Controls the size of the weight update step. Too high leads to divergence; too low slows convergence.
- **Batch Size:** Number of training samples processed before the model updates.
- **Number of Epochs:** How many times the model sees the full dataset.
- **Number of Hidden Layers & Neurons:** Network architecture design choices.
- **Activation Functions:** e.g., ReLU, Tanh, Sigmoid.
- **Optimizer:** Algorithm used to update weights — SGD, Adam, RMSProp, etc.
- **Dropout Rate:** Proportion of neurons randomly dropped during training.
- **Regularization Coefficients:** Values of λ in L1 or L2 regularization.

 Hyperparameter Tuning Methods

Method	Description	Pros	Cons
Grid Search	Tries all possible combinations of hyperparameter values	Thorough	Computationally expensive
Random Search	Randomly selects combinations from a predefined range	More efficient	May miss best combination
Bayesian Optimization	Uses previous results to model next best choice	Smart and adaptive	Requires more setup
Hyperband	Combines random search with early stopping	Fast and scalable	Can be complex to configure

 Tools for Tuning

- **Keras Tuner**
- **Optuna**
- **Ray Tune**
- **Hyperopt**

 Example:

Tuning a digit classifier (MNIST dataset):

- Grid search across number of layers (1–3), neurons (64, 128, 256), and learning rate (0.001–0.01).
- Use validation accuracy to select the best configuration.

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